RESEARCH PAPER ABSTRACT

Title: Deep Learning for Natural Language Processing: A Survey

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Abstract

This paper presents a comprehensive survey of deep learning techniques applied to natural language processing (NLP) tasks. The main research objective is to survey deep learning techniques applied to NLP tasks, covering architectures ranging from early recurrent neural networks to modern transformer-based models including BERT, GPT, and their derivatives. We analyse over 350 papers published between 2015 and 2023, evaluate performance across standard benchmarks, and identify open challenges. Our key finding is that transformer-based models consistently outperform classical approaches across all NLP benchmarks, while also requiring significantly greater computational resources. We discuss implications for both research and industry deployment, and propose directions for future work including efficiency improvements and multi-lingual generalisation.

Keywords: deep learning, natural language processing, transformers, BERT, GPT, survey

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